CE 4720 Microcomputer Architecture and Interfacing

Lab 3: Shift-and-Subtract Algorithm

Team Participation Requirement

Our lab work requires equal participation of all lab team members. When distributing the work on the lab assignments among the team members, make sure that everybody contributes to all stages of the assignment. Each member must contribute to the design and must have a portion of the Verilog code they can claim as the main authors, and all members must participate in running simulations, debugging as well as testing the final design on the DE2 boards. Similarly, all team members must actively participate in the preparation of the lab reports. Please read carefully Section 4 – Report – Appendix, which details how participation and authorship should be documented in the lab reports.

Not following these reporting requirements will result in notable point reduction for the team. Individuals reporting marginal contributions to the assignment will receive reduced points compared to the rest of the team. Please make sure to plan well and actively participate – that is the best way to learn and gain confidence and proficiency.

AI Use Policy

The use of AI/LLMs is permitted in lab assignments under the regulations defined in AI Use Policy in CE4720 (available on Canvas in the Supplementary Resources module).

1. Objective

- Understand basic sequential design constructs in Verilog.
- Practice implementation of an integer division algorithm.
- Practice the use of Quartus II software for creating, simulating, and synthesizing Verilog modules.

2. Task

Design a Verilog module that will implement the Shift-and-Subtract division algorithm for unsigned integers.

- The algorithm will perform the following function:

  $$Y:U \leftarrow UV \ / \ X$$

  where registers U and V will initially contain the dividend and X will contain the divisor. Upon the completion of the algorithm, Y will store the quotient and U the remainder. Implement U, V, X, Y as 4-bit registers.

- The implementation must follow the Shift-and-Subtract RTL code from the course lectures. Do not implement the division via the "/" Verilog operator and **do not use loop statements**. The system must be implemented as a state machine with the steps of the algorithms being states of the machine. For simulation purposes, generate a rectangular waveform that will be used as the clock signal. For the implementation on the DE2 board, use the onboard 50 MHz oscillator as the clock for the state machine.

- Use the DE2 switches to enter the input values and instances of the 7-segment decoder Verilog module from Lab 2 to display the result after the division.
- Dedicate one LED to display the Overflow status (no overflow → LED off, overflow → LED on). Use one of the DE2 push-buttons to produce the start signal. Study the DE2 manual for your board to understand the wiring of the push-buttons – do they use pull-up or pull-down resistors? Consider using a debouncing circuit to filter the transients from the push-button vibrating contacts.

3. Check-Off

Demonstrate the following divisions on the DE2 board. Convert the input values from decimal to binary for input via switches.

Test Cases

#	Division	Expected Result
1	`15 / 5`	Q = 3, R = 0
2	`180 / 14`	Q = 12, R = 12
3	`180 / 4`	Overflow

For the check-off videos, besides demonstrating the divisions, show the Verilog module(s) and briefly comment on how you approached the implementation (do not go over the code line-by-line).

4. Report

Follow the informal report format posted in the Canvas module. Each report should include the following:

Heading – the title of the lab, to whom the report is submitted, author list, the date it was submitted.

Summary – brief one or two paragraphs describing the objective and results.

Design:

- This section must detail all steps taken during the design of the system and its testing, including commented simulation results. The level of detail must be sufficient for an expert reader to be able to reproduce your design and evaluation procedure.
- Provide an RTL code for the Shift-and-Subtract algorithm together with a flowchart visualizing the algorithm (including the looping operation). Describe how you implemented these stages in Verilog, together with other aspects such as the sources of the input signals to your algorithm (the input values, the Start signal, clock).
- Include waveform simulations for all test cases from the Check-Off section. Annotate the waveform screenshots electronically in your text processor software to make it easy for the reader to understand which signal track represents what and where each of the three test cases start in the simulation. Present all test cases in a single waveform window.

Equipment List.

Conclusion and Discussion – extended version of the Summary with the team's subjective evaluation of the project, issues faced on the way, found workarounds, etc.

References.

Appendix:

- Well-commented Verilog source code; include a name of a student with each section of code they authored (in the comments).
- Summary of contributions – present a three-column table where the first column will list the team members, the second column their contributions (which sections of the Verilog code they wrote/co-wrote, involvement in other stages – simulation, debugging, testing on the board, report writing; note that ALL team members must equally participate in the design stage and report writing) and the third column will present the total time spent by each member on the project.

5. Submission

Lab report in PDF format; submit one report per team (include names of all team members).

6. Dates

- Check-off due date is Thursday, April 23.
- Lab report due: Monday, April 27, 11:59 pm.

7. Student Checklist for Report & Deliverables

Use this checklist to make sure your Lab 3 submission meets all requirements. Each item may be graded, and missing pieces can cost points.

Category	Checklist Items
Division Algorithm Implementation	☐ Shift-and-Subtract algorithm implemented following the RTL code from lectures.
	☐ U, V, X, Y implemented as 4-bit registers.
	☐ Division is not implemented via the "/" Verilog operator.
	☐ Algorithm correctly computes quotient (Y) and remainder (U) upon completion.
	☐ Overflow condition is correctly detected and signaled.
I/O & Board Integration	☐ DE2 switches used to enter dividend (UV) and divisor (X).
	☐ 7-segment decoder modules from Lab 2 used to display results.
	☐ Dedicated LED displays overflow status (off = no overflow, on = overflow).
	☐ Push-button used as the start signal; pull-up/pull-down behavior addressed.
	☐ Debouncing circuit considered or implemented for the push-button.
	☐ Correct pin assignments used for DE2 board version (Cyclone II or Cyclone IV).
Simulation & Testing	☐ Quartus II simulation run for all three check-off test cases (15/5, 180/14, 180/4).
	☐ Waveform screenshots included in Design section with annotations.
	☐ All three test cases presented in a single waveform window.
	☐ Signals clearly labeled and test case boundaries indicated.
	☐ Simulation results show correct quotient, remainder, and overflow for all test cases.
Check-Off Readiness	☐ Design successfully synthesized and tested on DE2 board.
	☐ Can demonstrate all three test cases on the DE2 board.
	☐ Can convert decimal inputs to binary for switch input.
	☐ Can verify displayed results match expected quotient, remainder, and overflow.
	☐ Can show Verilog module(s) and explain the implementation approach.
	☐ Can explain the Shift-and-Subtract algorithm stages and RTL mapping.
Report Structure & Quality	☐ Report follows the informal report template (Heading, Summary, Design, Equipment List, Conclusion/Discussion, References, Appendix).

	☐ Summary states the objective and key results.
	☐ Design section describes approach clearly enough for reproduction without seeing the code.
	☐ RTL code for the Shift-and-Subtract algorithm is included with a flowchart.
	☐ Description of how RTL stages were implemented in Verilog is provided.
	☐ Sources of input signals (input values, Start signal) are described.
	☐ Waveform screenshots include all three test cases with annotations.
	☐ Screenshots are clear, readable, and referenced in the text.
	☐ References section lists any external sources used.
Appendix & Team Contributions	☐ Appendix includes complete, well-commented Verilog source code.
	☐ Code comments indicate student authorship for major sections.
	☐ Appendix includes a contributions table: Team member | Contributions | Total time spent.
	☐ Contributions indicate that all members participated in design, simulation/debugging, and report writing.
AI Use Disclosure (if applicable)	☐ AI/LLM use follows the CE4720 AI Use Policy posted on Canvas.
	☐ If an AI tool was used, the report clearly discloses it (tool name + what it was used for).
	☐ AI was not used to produce large copy-pasted blocks of final Verilog code or report text.
Final Submission	☐ One PDF report submitted to Canvas by the due date (one per team, with all team member names).
	☐ Check-off completed by the due date (in-person or approved video process).
	☐ File names are sensible and match course expectations.